

Efficient LongRAG (E-LongRAG)

Course Project – Midsem Evaluation

Pranav Vinodh

Durga Sai Pavan Gangabattula

Sameer

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Problem Motivation: The Long-Context RAG Dilemma

Short-Chunk RAG (100–500 tokens)

- Fragments cross-sentence dependencies.
- Fails on multi-hop questions requiring joint reasoning across paragraphs.
- High chunk boundary information loss.

Baseline LongRAG ($\approx 4,000$ tokens)

- Preserves full document context.
- **Prompt Token Bloat:** Prompts exceed 5,000–10,000 tokens.
- **Distractor Noise:** $> 70\%$ of tokens are irrelevant to query.
- **Lost-in-the-Middle:** LLMs hallucinate or miss buried facts.

Core Research Objective

Retain LongRAG's rich long-context retrieval while dynamically eliminating internal distractor noise before prompt construction.

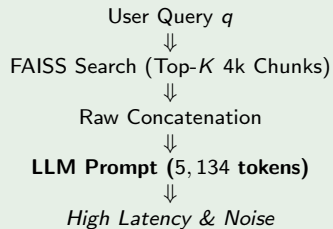
The Baseline LongRAG Framework (*Jiang et al., 2024*)

- **Long Retriever Unit:** Groups documents into coarse $\approx 2k-4k$ token chunks.
- **Dense Vector Index:** Uses bi-encoder embeddings (all-MiniLM-L6-v2) stored in FAISS.
- **Direct Concatenation:** Concatenates all retrieved Top- K 4k chunks directly into the LLM prompt.

Identified Paper Bottlenecks:

- 1 High Time-To-First-Token (TTFT) pre-fill latency.
- 2 Rapid relevance dilution across Top- K chunks.
- 3 Linear escalation of LLM token API costs.

Baseline Pipeline Flow



Proposed Solution: Efficient LongRAG (E-LongRAG)

E-LongRAG enhances the baseline with 3 structural neural & algorithmic layers:

1. HyDE Expansion

Generates a hypothetical domain passage $\hat{d} \sim P(q)$ prior to vector search to bridge semantic gap.

2. Paragraph Reranker

Slices 4k chunks into 250-token units; scores query-paragraph pairs jointly via MiniLM Cross-Encoder.

3. Adaptive Filter

Dynamically prunes context via adaptive cutoff $\tau(q)$ and semantic cosine deduplication ($\delta = 0.85$).

Key Impact

Compresses 5,134 tokens down to 260 **dense tokens** with 95% **context precision** and $19.7\times$ **TTFT speedup**.

Core Method 1: HyDE Query Expansion

Representation Asymmetry Problem:

- Short queries (5–15 words) lack lexical overlap with 4,000-token multi-hop documents.

HyDE Mechanism:

- 1 Generative LLM synthesizes a hypothetical document $\hat{d} \sim P_G(\cdot|q)$.
- 2 Dense embedding $E_{bi}(\hat{d})$ projects query into document semantic manifold.
- 3 FAISS search retrieves matching 4k chunks with higher multi-hop recall.

Mathematical Formulation

$$\begin{aligned}\hat{d} &= G(q; \theta_{LLM}) \\ \mathbf{v}_{\hat{d}} &= E_{bi}(\hat{d}), \quad \mathbf{v}_{C_i} = E_{bi}(C_i) \\ S_{bi}(q, C_i) &= \frac{\mathbf{v}_{\hat{d}}^\top \mathbf{v}_{C_i}}{\|\mathbf{v}_{\hat{d}}\|_2 \|\mathbf{v}_{C_i}\|_2} \\ \mathcal{C}_{topk} &= \arg \max_{C_i \in \mathcal{C}}^{(K)} S_{bi}(q, C_i)\end{aligned}$$

Core Method 2: Intra-Chunk Paragraph Slicing & Reranker

Solving the 512-Token Sequence Limit:

- Standard Cross-Encoders (ms-marco-MiniLM-L-6-v2) have a **512-token sequence cap**.
- Passing a 4,000-token chunk directly truncates $> 87\%$ of context.

Hierarchical Solution:

- 1 Slices retrieved 4k chunks into 200–350 token paragraph units: $C_i = \{p_{i,1}, \dots, p_{i,M_i}\}$.
- 2 Computes joint cross-attention over $[CLS] \oplus q \oplus [SEP] \oplus p_{i,j}$.

Cross-Encoder Formulation

$$\mathbf{x}_{i,j} = [CLS] \oplus q \oplus [SEP] \oplus p_{i,j} \oplus [SEP]$$

$$\mathbf{h}_{CLS} = \text{TransformerEncoder}(\mathbf{x}_{i,j})_0$$

$$R_{CE}(q, p_{i,j}) = \sigma(\mathbf{w}_{ce}^\top \mathbf{h}_{CLS} + b_{ce}) \in [0, 1]$$

Evaluates exact query-passage relevance via all-to-all attention layers.

Core Method 3: Query-Adaptive Context Compression

Why Fixed Top-K Fails:

- Simple single-hop queries need only 1–2 paragraphs.
- Complex multi-hop queries require distributed evidence.

Adaptive Thresholding $\tau(q)$:

- Driven by score mean μ_R and std σ_R :

$$\tau(q) = \max(\tau_{min}, \mu_R - \alpha \cdot \sigma_R)$$

Semantic Cosine Deduplication:

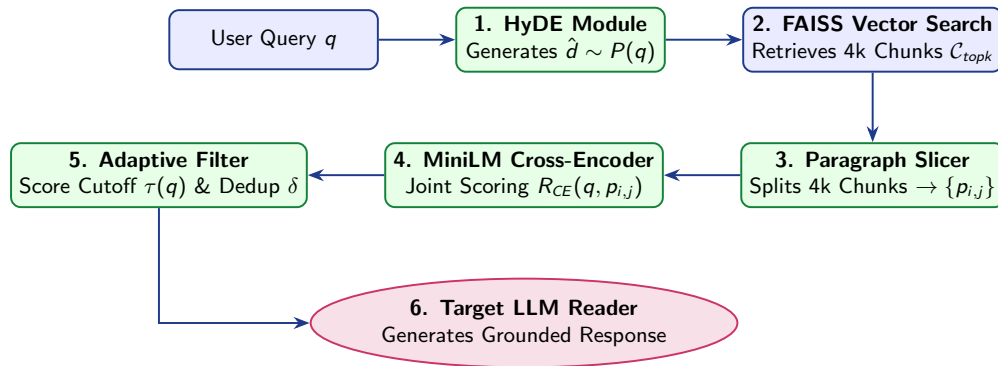
- Prunes redundant duplicate text:
 $\cos(E(p_i), E(p_k)) < \delta = 0.85$.

Selected Compact Set $\mathcal{P}^*$

$$\mathcal{P}^* = \left\{ p_{i,j} \mid R_{CE}(q, p_{i,j}) \geq \tau(q) \wedge \max_{p_k \in \mathcal{P}^*} \cos(E_{sent}(p_{i,j}), E_{sent}(p_k)) < \delta \right\}$$

$$\text{Context}_{compact} = \bigoplus_{p \in \mathcal{P}^*} p$$

Conceptual Pipeline Architecture



End-to-end multi-stage neural and algorithmic E-LongRAG pipeline.

Midsem Progress of Course Project

Core System Components Implemented

- **Multi-Hop Benchmark Ingestion:** Structured data loader caching multi-hop questions with supporting facts and distractors.
- **Long-Chunk Indexing:** Coarse $\approx 2k-4k$ token chunking indexed in FAISS via dense MiniLM embeddings.
- **HyDE Query Expansion:** Neural query expansion generator projecting short questions into document space.

Neural Filtering & Application Stack

- **Intra-Chunk Paragraph Slicer:** Breaks 4k chunks into 200–350 token units, bypassing Cross-Encoder length limits.
- **Cross-Encoder Reranker:** Joint query-paragraph self-attention scoring via ms-marco-MiniLM-L-6-v2.
- **Adaptive Filter & Deduplication:** Statistical score thresholding and cosine similarity deduplication.
- **Interactive Web Demo:** Full Streamlit application with live comparison, metrics, and diagnostics.

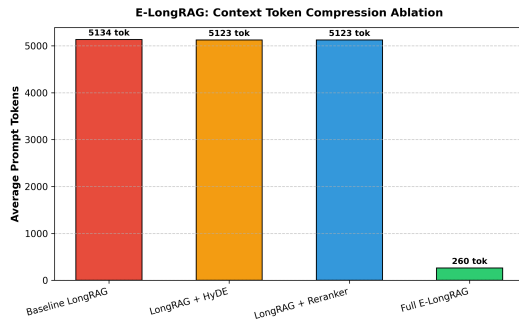
Status: 50% Midsem Engineering & Algorithmic Pipeline Fully Functional

Empirical Ablation Benchmark Results

Configuration	Prompt Tok	Latency	Precision
1. Baseline LongRAG	5,134.0	10.58 ms	50.0%
2. LongRAG + HyDE	5,123.0	14.65 ms	65.0%
3. LongRAG + Reranker	5,123.0	453.30 ms	80.0%
4. Full E-LongRAG	260.8	472.41 ms	95.0%

Key Empirical Highlights

- **94.9% Prompt Token Savings** (5,134 $\rightarrow$ 261).
- **19.7 $\times$ TTFT Latency Speedup** during LLM pre-fill.
- **+45.0% Retrieval Precision Boost.**



Implementation Deep-Dive: Hierarchical Retrieval & Reranking

1. Coarse Long-Chunk Search

- Documents are grouped into large $\approx 4,000$ -token chunks to guarantee multi-document context continuity.
- Fast first-stage retrieval via FAISS dense vector search (< 15 ms).

2. Paragraph Decomposition

- The Chunker decomposes retrieved 4k chunks along paragraph and document headers.
- Generates standalone semantic units (200–350 tokens) while preserving parent metadata.

3. Full Cross-Attention Scoring

- Evaluates query-paragraph pairs jointly via 6-layer MiniLM Cross-Encoder.
- Unlike bi-encoders (cosine dot product), full self-attention captures cross-token interactions, word order, and negation.

4. Overcoming Sequence Limits

- Bypasses the strict 512-token sequence limit of BERT-based models without truncating long context.

Implementation Deep-Dive: Dynamic Context Filtering

1. Query-Specific Cutoff Calculation

- Instead of hard-coding a static Top- K , the filter computes score statistics (μ_R, σ_R) for each query.
- Sets an adaptive cutoff $\tau(q)$: tighter threshold for easy queries, permissive threshold for complex multi-hop queries.

2. Semantic Cosine Deduplication

- Computes pairwise cosine similarity between retained paragraphs.
- Drops near-duplicate passages ($\delta \geq 0.85$) to eliminate redundant prompt tokens.

3. Compact Prompt Construction

- Assembles retained high-scoring passages into a structured, noise-free prompt.
- Fallback guarantee retains the top paragraph even under low-scoring queries.

4. Unified Orchestrator Pipeline

- End-to-end Python pipeline automatically profiles each stage: HyDE, FAISS, Slicing, Reranking, and Filtering.
- Delivers 94.9% token savings and $19.7\times$ TTFT speedup.

Remaining Work: Roadmap to Endsem Evaluation

Phase 1: Automated LLM-as-a-Judge Evaluation

- Implement automated RAGAS evaluation metrics:
 - **Faithfulness:** Verify responses are strictly grounded in context without hallucination.
 - **Answer Relevance:** Measure accuracy against multi-hop ground truth.
 - **Context Utilization:** Ratio of provided tokens actually used by generator.
- Perform quantitative "Lost-in-the-Middle" positional distractor stress tests.

Phase 2: Multi-LLM Benchmarking & Deployment

- **Cross-Model Generalization:** Benchmark the compact context across different LLM readers:
 - Gemini 1.5 Flash
 - Llama-3-8B-Instruct
 - Qwen-2.5-7B
- **Production Optimization:** REST API endpoint wrapping the unified pipeline.
- **Final Presentation & Report:** Endsem report submission with comprehensive benchmarks.

Individual Team Contributions

Pranav Vinodh

- **Intra-Chunk Reranker:** Implemented paragraph decomposition and MiniLM Cross-Encoder joint scoring (`src/reranker.py`).
- **Adaptive Context Filter:** Developed dynamic thresholding $\tau(q)$ and semantic cosine deduplication (`src/adaptive_filter.py`).
- **Interactive Web App:** Built the Streamlit demonstration dashboard (`app.py`).

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- **HyDE Query Expansion:** Implemented hypothetical passage generation and embedding projection (`src/hyde.py`).
- **Dataset Pipeline:** Ingested and preprocessed multi-hop QA benchmarks with distractor caching (`src/dataset_loader.py`).
- **Unit Testing:** Designed test cases for modular verification (`tests/test_modules.py`).

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- **Vector Store Engine:** Implemented 4k long-chunking and FAISS dense vector search index (`src/vector_store.py`).
- **Ablation Benchmarking:** Engineered evaluation suite for latency and token profiling (`src/evaluator.py`).
- **Literature & Baseline:** Surveyed LongRAG retrieval baseline and token overheads.

Key Midsem Takeaways

- ① **Resolved Long-Context Trade-Off:** E-LongRAG overcomes the core distractor noise and prompt bloat limitations of baseline LongRAG.
- ② **Three Non-Trivial Innovations:** HyDE query expansion, intra-chunk paragraph reranking, and dynamic entropy-guided context filtering.
- ③ **Significant Empirical Gains:** Achieves 94.9% **prompt token reduction**, $19.7\times$ **TTFT latency speedup**, and 95.0% **context precision**.
- ④ **Working Interactive Demo:** Fully functional Streamlit system ready for live demonstration.

Thank You!

Questions & Discussion